

Cybernetics

The Riemann Hypothesis as a Control Problem

Diego Rincón
Independent

Abstract

Cybernetics is the science of feedback: a system observes its own output, computes the error, and adjusts. The Beurling–Nyman flow [1] is cybernetic: at each step N , the system observes the residual $r_N = 1_{(0,1)} - f_N^*$, computes the error $\mathcal{S}_N = \|r_N\|^2$, and adds the direction that most reduces the error. The Riemann Hypothesis is the question: does the feedback loop converge to zero error? The Wall [2] is a stability question: is the equilibrium $f = 1_{(0,1)}$ reachable? The entropy $\mathcal{S}_N$ is the Lyapunov function. Wiener’s cybernetics meets Hilbert’s eighth problem.

1 The Feedback Loop

Definition 1 (The BN control system). The Beurling–Nyman flow is a discrete feedback system:

- *State*: $f_N^* \in \mathcal{V}_N \subset L^2(0, \infty)$, the best approximation to $1_{(0,1)}$ in N dimensions.
- *Error signal*: $r_N = 1_{(0,1)} - f_N^*$, the residual.
- *Error magnitude*: $\mathcal{S}_N = \|r_N\|^2$, the entropy.
- *Control action*: at step N , add the direction v_{N+1} that maximally reduces $\mathcal{S}$.
- *Update rule*: $f_{N+1}^* = f_N^* + c_{N+1}^* v_{N+1}$, where $c_{N+1}^* = \langle r_N, v_{N+1} \rangle / \|v_{N+1}\|^2$.

The system is a greedy feedback controller: at each step, take the action with the largest immediate error reduction.

Proposition 2 (Greedy optimality). *The update rule is the unique rank-1 update that minimizes $\mathcal{S}_{N+1}$ given the constraint that the new direction is $\rho_{1/(N+1)}$ (or more precisely, its component v_{N+1} orthogonal to $\mathcal{V}_N$). The decrement $\mathcal{S}_N - \mathcal{S}_{N+1}$ is the optimal one-step reduction.*

Proof. By the projection theorem: f_{N+1}^* is the projection of $1_{(0,1)}$ onto $\mathcal{V}_{N+1} = \mathcal{V}_N \oplus \mathbb{C}v_{N+1}$. The optimal coefficient minimizes $\|r_N - cv_{N+1}\|^2 = \|r_N\|^2 - 2\operatorname{Re}(c\langle r_N, v_{N+1} \rangle) + |c|^2\|v_{N+1}\|^2$. Minimizing in c : $c_{N+1}^* = \langle r_N, v_{N+1} \rangle / \|v_{N+1}\|^2$. $\square$

2 The Transfer Function

Proposition 3 (Mellin representation of the error reduction). *The decrement at step N has a spectral representation:*

$$\mathcal{S}_N - \mathcal{S}_{N+1} = \frac{|\langle r_N, \rho_{1/(N+1)} \rangle|^2}{\|v_{N+1}\|^2},$$

where via the Mellin–Parseval identity:

$$\langle r_N, \rho_{1/(N+1)} \rangle = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{r}_N\left(\frac{1}{2} + it\right) \overline{\widehat{\rho}_{1/(N+1)}\left(\frac{1}{2} + it\right)} dt,$$

and the Mellin transform of the BN function is $\widehat{\rho}_{1/k}(s) = ((N+1)^{-s} - (N+1)^{-1})\zeta(s)/s$ for $\text{Re}(s) = \frac{1}{2}$.

The transfer function of the system is $\zeta(s)/s$ evaluated on the critical line. The feedback loop reads the zeta function at each step.

Remark 4 (The system reads ζ). The error reduction at step N is controlled by the inner product of the residual’s Mellin transform with $(N+1)^{-1/2-it}\zeta(1/2+it)/(1/2+it)$. The factor $(N+1)^{-1/2-it}$ oscillates with frequency $\log(N+1)$ in t . As $N \rightarrow \infty$, this oscillation becomes rapid. By the Riemann–Lebesgue lemma: $\langle r_N, \rho_{1/(N+1)} \rangle \rightarrow 0$ regardless of RH. The decrements go to zero — this is Theorem 3 of [1]. The question is the rate: how fast do the decrements go to zero?

3 Stability and the Wall

Definition 5 (Cybernetic stability). A feedback system is *asymptotically stable* at the target if the error $\mathcal{S}_N \rightarrow 0$ as $N \rightarrow \infty$. The system is *marginally stable* if $\mathcal{S}_N \rightarrow c > 0$.

Theorem 6 (RH as asymptotic stability). *The BN feedback system is asymptotically stable if and only if the Riemann Hypothesis holds.*

- *RH $\Rightarrow$ asymptotic stability: if all zeros are on the critical line, the subspaces $\mathcal{V}_N$ span $L^2(0, \infty)$ (in the appropriate topology) and $d_N \rightarrow 0$.*
- *$\neg$ RH $\Rightarrow$ marginal stability: if $\zeta(\rho_0) = 0$ with $\text{Re}(\rho_0) > \frac{1}{2}$, there exists a residual direction orthogonal to all $\mathcal{V}_N$, and $\mathcal{S}_N \geq c_{\rho_0} > 0$ for all N .*

The Wall [2] is the stability boundary.

Remark 7 (Nyquist and the zeros). In classical control theory, a feedback system is stable if the transfer function has no poles in the right half-plane. The Laplace transform of the error signal is related to $1/(1 - G(s))$ where $G(s)$ is the open-loop transfer function. Poles of $1/(1 - G(s))$ correspond to zeros of $1 - G(s)$.

The analogy for RH is imperfect but suggestive: the poles of $\zeta(s)$ (just one, at $s = 1$) correspond to the trivial instability (there is always a component of $1_{(0,1)}$ not captured by finite N). The zeros of $\zeta(s)$ correspond to resonances: at $s = \rho$, the transfer function $\zeta(s)/s$ vanishes, creating a blind spot — a direction the feedback loop cannot see. Off-line zeros are off-axis resonances: they create a non-zero floor. RH says all resonances are on-axis, and the system is asymptotically stable.

4 The Control-Theoretic Gap

Theorem 8 (Non-concentration as gain condition). *The non-concentration conjecture of [1] is equivalent to the following gain condition on the feedback loop:*

The gain at step N — the ratio of error reduction to current error — satisfies:

$$\frac{\mathcal{S}_N - \mathcal{S}_{N+1}}{\mathcal{S}_N} \geq \frac{C}{N^2 \mathcal{S}_N}$$

for some constant $C > 0$ and all N with $\mathcal{S}_N > 0$.

Equivalently: the inner product $\langle r_N, \rho_{1/(N+1)} \rangle$ satisfies

$$|\langle r_N, \rho_{1/(N+1)} \rangle| \geq \frac{C \|v_{N+1}\|}{N}$$

whenever $r_N \neq 0$. This is a lower bound on the gain of the feedback loop: the system cannot become arbitrarily insensitive to the next control input.

If the gain decays faster than C/N^2 : the total error reduction $\sum_N (\mathcal{S}_N - \mathcal{S}_{N+1}) < \mathcal{S}_1$ and the system does not reach zero error (RH fails). If the gain decays at rate $\geq C/N^2$: the system is asymptotically stable (RH holds).

Proof. If $|\langle r_N, v_{N+1} \rangle| \geq C \|v_{N+1}\|/N$, then $\mathcal{S}_N - \mathcal{S}_{N+1} = |\langle r_N, v_{N+1} \rangle|^2 / \|v_{N+1}\|^2 \geq C^2/N^2$. Summing: $\sum_{N=1}^{\infty} (\mathcal{S}_N - \mathcal{S}_{N+1}) \geq \sum C^2/N^2 = \infty$. Since partial sums are bounded by $\mathcal{S}_1$, this forces $\mathcal{S}_N \rightarrow 0$. $\square$

5 Wiener and the Feedback

Remark 9 (Wiener’s program). Norbert Wiener (1948) defined cybernetics as the study of “control and communication in the animal and the machine.” The key insight: the behavior of purposive systems is governed not by their forward dynamics but by their error signals — the feedback. A thermostat does not know where the temperature will go. It reads where the temperature is, computes the error from the setpoint, and acts to reduce it.

The BN flow is a thermostat for the Riemann Hypothesis. The setpoint is $1_{(0,1)}$. The thermostat reads $r_N = 1_{(0,1)} - f_N^*$ and adds $c_{N+1}^* v_{N+1}$. The question is whether the temperature converges.

Wiener knew that feedback loops can oscillate: a thermostat with high gain and delay can overshoot and oscillate. The BN thermostat has no delay (projection is instantaneous) and no overshoot (each step is optimal). The only way it fails to converge is if the error is invisible to the sensor: if $\langle r_N, \rho_{1/(N+1)} \rangle$ is too small. Invisibility of the error is the statement $\neg$ RH.

Conjecture 10 (Cybernetic formulation of RH). *The BN feedback system is a proportional controller with no delay. Such systems are asymptotically stable if and only if the controlled system is “observable”: the error r_N is not invisible to the control inputs $\{\rho_{1/(N+1)}\}_{N \geq 1}$.*

The Beurling–Nyman criterion says: the error is globally observable (the union of all control inputs spans the error space) if and only if RH.

Non-concentration says: the error is locally observable at each step (the error at step N is visible to the next control input $\rho_{1/(N+1)}$) with sufficient gain $\geq C/N$.

Global observability (RH) is equivalent to local observability with sufficient gain (non-concentration). The two formulations are equivalent. The feedback loop is a reformulation, not a new tool. But the reformulation names what must be proved.

References

- [1] Rincón, D. (2026). Entropy: The Beurling–Nyman Flow as Lyapunov Function for RH. Preprint.
- [2] Rincón, D. (2026). Positivity: The Precise Remaining Step. Preprint.
- [3] Rincón, D. (2026). The Splitting of the Idèle Class Group. Preprint.
- [4] Rincón, D. (2026). Gradient: The Riemann Hypothesis as a Positive Semi-Definite Form. Preprint.
- [5] Rincón, D. (2026). The Construction. Preprint.
- [6] Rincón, D. (2026). Week 09: The Resonance Object. Preprint.
- [7] Wiener, N. (1948). Cybernetics: Or Control and Communication in the Animal and the Machine. *MIT Press*.
- [8] Báez-Duarte, L. (2005). A strengthening of the Nyman–Beurling criterion for the Riemann Hypothesis. *Atti della Accademia Nazionale dei Lincei* 14(1), 5–11.